Electronic supplementary material 1. GRADE evidence profile

Summary of findings				Quality assessment					
Outcome	No. of participants (studies)	Pooled treatment effect (95% CI)	I^2	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias†	Quality rating
Body mass (kg)	660 (4)	-1.6 (-2.7, -0.39)	81%	Serious limitations[a]	Serious inconsistency[b]	No serious indirectness	No serious imprecision[c]	Undetected	Low
Body mass index (kg/m^2)	395 (3)	-0.78 (-1.1, -0.50)	21%	Serious limitations[d]	No serious inconsistency	No serious indirectness	No serious imprecision[c]	Undetected	Moderate
Waist circumference (cm)	392 (3)	-2.9 (-3.8, -1.9)	0%	Serious limitations[d]	No serious inconsistency	No serious indirectness	No serious imprecision[c]	Undetected	Moderate
Physical activity	440 (4)	0.31 (0.13, 0.50)	0%	Serious limitations[a]	No serious inconsistency	Serious indirectness[e]	No serious imprecision[c]	Undetected	Low
Fruit and vegetable intake	432 (4)	0.33 (0.01, 0.63)	51%	Serious limitations[a]	Serious inconsistency[f]	No serious indirectness	No serious imprecision[c]	Undetected	Low
Fibre intake	432 (3)	4.3 (-3.0, 11.5)	92%	Serious limitations[a]	No serious inconsistency[g]	No serious indirectness	Serious imprecision[h]	Undetected	Low

[a]Most of the evidence comes from studies with a crucial limitation in one domain of Cochrane risk of bias tool for randomized trials (RoB 2) or some limitations in multiple domains.

[b]The proportion of between-study variance due to heterogeneity was considerable and removal of individual studies (via a Leave-One-Out analysis) influenced the meta-analysis results so that the 95% CI crossed the line of no effect.

[c]The total number of participants exceeds to Optimal Information Size (OIS)* criterion and the 95% confidence interval (CI) excludes zero, and therefore the evidence was not downgraded.

[d]Most of the evidence comes from studies with a crucial limitation in one domain of RoB 2.

[e]Measures of physical activity included both objective and self-reported measures. Self-reported measures may be suboptimal and are likely to be prone to response bias compared with objectively-measured physical activity.

[f]The proportion of between-study variance due to heterogeneity was moderate and removal of individual studies (via a Leave-One-Out analysis) influenced the meta-analysis results so that the 95% CI crossed the line of no effect.

[g]The proportion of between-study variance due to heterogeneity was considerable, however, removal of one study reduced I^2 to 0%. Therefore, we did not downgrade the evidence because the considerable heterogeneity was entirely explained by one study.

[h]The total number of participants exceeds to OIS criterion, but the 95% CI crossed zero and the upper CI bound failed to exclude the minimum clinically important difference (MCID; considered to be 5 arbitrary units).

*For mean differences, the OIS was calculated using $\alpha = 0.05$, $\beta = 0.20$, an independent *t*-test, and by identifying an MCID and extracting the SDs associated with the relevant outcome from studies included in the meta-analysis (see Table below). For standardised mean differences, the minimum important difference for calculation of the OIS was 0.5 SDs.
†The low number of studies precluded the assessment of publication bias for each outcome using funnel plot or regression-based methods. Therefore, we combined all effect estimates across all outcomes in one funnel plot by calculating standardised mean differences (and corresponding sampling variances) and nesting effect sizes within each study. As a result, the assessment of publication bias relates to all outcomes included in the review, not each individual outcome.

Minimum clinically important difference (MCID) and standard deviations (SDs) used to inform calculation of the Optimal Information Size (OIS) for outcomes reported as a mean difference

Outcome	MCID	Most conservative SD from relevant studies	OIS
Body mass (kg)	2.5 kg	4.5 kg	106
Body mass index (kg/m^2)	0.5 kg/m^2	1.5 kg/m^2	292
Waist circumference (cm)	3 cm	6.0 cm	128
Fibre intake	5 arbitrary units	8.7 arbitrary units	100